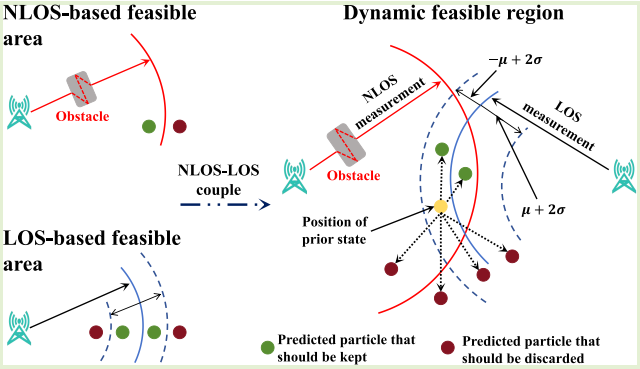
Dynamic Feasible Region-Based IMU/UWB Fusion Method for Indoor Positioning

Jiong Liu, Li Zhang, Jingao Xu, *Member, IEEE*, and Jun Shi

***Abstract*—Given that the challenges inherent in complex indoor environments where GPS signals are often weak, various alternative signals have been employed for indoor positioning. Ultra-wideband (UWB) technology, distinguished by its high-precision positioning capability, has an extensive application in fields of indoor positioning. Nevertheless, it is susceptible to the effects of multipath propagation and nonline-of-sight (NLOS) conditions, which can introduce significant inaccuracies into positioning outcomes. To improve the accuracy and stability of positioning in complex indoor environments, this study presents a dynamic feasible region-based particle filter (DFRPF). It effectively solves the issues of particle convergence and weak robustness to NLOS. Moreover, the observation likelihood function undergoes dynamic adjustment based on the received UWB signal power quality, thereby enhancing the system's adaptability to diverse environments. Numerical experiments in different scenarios show that the proposed fusion approach can maintain high levels of accuracy and stability in complex indoor environments.**



***Index Terms*—Indoor positioning, inertial measurement unit (IMU), nonline-of-sight (NLOS), particle filter (PF), ultra-wideband (UWB).**

I. Introduction

AMONG the swift progress of Internet of Things (IoT), there has been a growing interest in location services [1], [2], [3]. Attaining high-precision and stable positioning outdoors is feasible with the global positioning system (GPS) [4], but achieving the same level of accuracy indoors remains a great challenge. Therefore, different indoor positioning system (IPS) technologies have been proposed and developed to address this challenge. These technologies encompass Wi-Fi [5], [6], radio frequency identification (RFID) [7], [8], Bluetooth low energy (BLE) [9], [10], and ZigBee [11]. The accuracy of the positioning methods mentioned above remains insufficient for precise indoor positioning. In order to meet the requirements of high-precision pedestrian navigation, UWB has been widely used in recent years due to its low power consumption, high data rates, precise ranging, and robust immunity to interference ability [12], [13]. Meanwhile, numerous methods based on the UWB have been proposed for indoor positioning. For instance, the least square (LS) positioning algorithm based on the UWB measured distances [14]. In [15], a pedestrian tracking system based on the fusion of inertial navigation system (INS) and UWB has been proposed. To reduce the errors caused by clock drift, asynchronous hyperbolic localization algorithm is proposed [16]. These approaches can reduce the ranging errors to an acceptable level in ideal conditions. However, in a challenging indoor environment, obstacles like walls or pedestrians can extend signal transmission distance and delay transmission time, thereby significantly compromising positioning accuracy. This phenomenon is commonly referred to as a nonline-of-sight (NLOS) case [17], [18], [19]. Typically, the uncertainty observed in UWB ranging measurements can be modeled using an error model that incorporates a Gaussian distribution for LOS measurements, a Gamma distribution for NLOS measurements, and a constant value [20]. To reduce localization errors caused by NLOS, it is essential to identify NLOS condition and eliminate their impact. Numerous approaches for identifying NLOS condition have been suggested. These methods include but not limited to the residual analysis judgment methods [21], analysis based on the channel impulse response (CIR) [22], and machine learning methods [23].

To enhance the accuracy of localization, multisensor fusion algorithms are typically required to leverage the advantages of each sensor to mitigate the adverse effects of NLOS. Common techniques for this purpose include unscented Kalman filter

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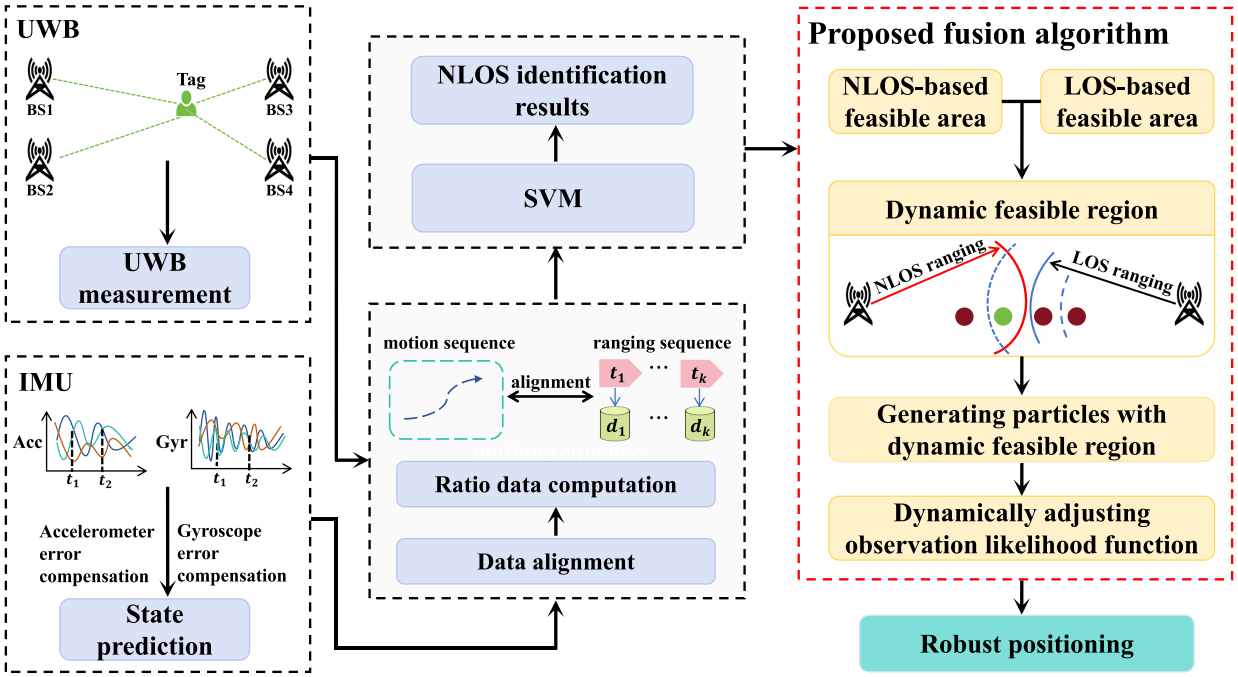


Fig. 1. System overview.

(UKF) [24], extended Kalman filter (EKF) [25], unbiased finite impulse response filtering (UFIR) [26], and particle filter (PF) [27]. Among these, the PF stands out for its ability to effectively handle non-Gaussian systems and track multimodal probability distributions. Therefore, it has been widely used in positioning systems and has shown excellent performance. However, it is important to note that the PF may encounter challenges such as particle degeneracy and limited robustness. As a result, numerous researchers have endeavored to enhance the robustness of traditional particle filtering algorithms through various improvements. In [28], techniques are introduced that individually integrate bitmap or standard floor plans into particle filters. The goal is to remove particles that appear in impossible regions during propagation, thereby optimizing particle distribution and improving positioning accuracy. However, maintaining the accuracy of system requires prompt updating of the bitmap or standard floor plan when alterations occur in the indoor environment, such as modifying room layouts or introducing obstacles. This process demands additional manpower and time resources to adapt to environmental changes. Therefore, in this article, we propose a dynamic feasible region-based particle filter (DFRPF), which incorporates a dynamic feasible region-based particle sampling (DFRPS) methodology. It solely leverages the identified LOS and NLOS range measurements to define a dynamic feasible region (DFR) for particle generation, thereby solving the problem of particle degeneracy. Moreover, since the proposed method is independent of environment factors, it achieves greater adaptability and robustness. In addition, since different UWB states have distinct ranging error models, the observation likelihood function is dynamically adjusted based on the received UWB signal power quality in response to changing environments. The main contributions can be summarized as follows.

1) We propose a DFRPS methodology. The DFRPS generates particles in collaboration with the DFR, rather than conducting particles resampling following the state transition. Furthermore, it initializes the weight of each predicted particle based on the angular disparity. The methodology effectively addresses the problem of particle convergence and weak robustness to NLOS.
2) In the importance weighting stage of DFRPF, we dynamically adjust the observation likelihood function based on the differences in UWB measurement states. Compared to traditional importance weighting, this method improves the adaptability and positioning accuracy of system across changing environments.
3) We conduct experiments in two distinct scenarios to validate our algorithm. The experimental results demonstrate that our proposed algorithm has the ability to copy with the dynamic and complex indoor environments.

II. Building Modules of the Proposed Positioning System

In this section, we introduce the proposed fusion positioning framework and provide an overview of the key sensor modules of the system.

As depicted in Fig. 1, the proposed algorithm primarily integrates data from the IMU and UWB sensors. The IMU sensors, consisting of a three-axis accelerometer and a three-axis gyroscope, can provide the direction and acceleration information. The UWB sensors, including a tag with an unknown position and four base stations (BSs) with known positions, are utilized for ranging purposes. NLOS identification is accomplished through the implementation of an SVM-based method. Subsequently, leveraging the NLOS identification results, the fusion algorithm defines a rational DFR for particle generation

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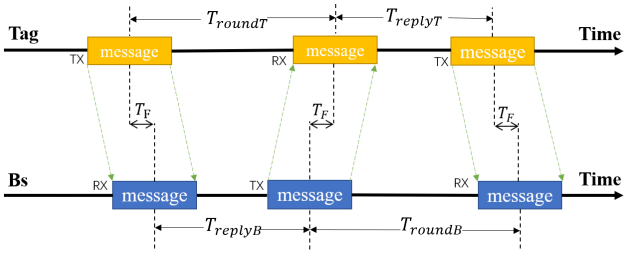


Fig. 2. SDS-TWR protocol.

and dynamically adjusts the observed likelihood function to account for variations in the environment.

A. UWB Module

UWB technology is a wireless communication technology distinguished by its utilization of a wide spectrum of radio frequencies for the transmission of data through short, high-frequency pulses. In contrast to conventional wireless communication methodologies reliant on narrow frequency bands, UWB broadcasts its signal across an extensive frequency range, typically spanning multiple gigahertz.

In our system, UWB sensors are utilized for ranging purposes. Through the exchange of information between two nodes, the time of flight (TOF) can be calculated to determine distance.

Fig. 2 illustrates the message-exchange procedure of the symmetric double sided two way ranging (SDS-TWR) methodology. The procedure initiates with the transmission of a message packet from the tag to the BS. Following a flight time denoted as T_F, the BS receives the transmitted packet. Subsequently, a substantial processing interval is allocated for packet handling and the generation of an acknowledgment packet, referred to as T_{replyB}. Upon the reception of this acknowledgment packet, the round-trip time T_{roundT} is determined by the tag. The tag then transmits another message, incorporating a time delay parameter identified as T_{replyT}. Upon arrival at the BS's antenna, this message enables the calculation of round-trip time T_{roundB}. So, we can get

$$\begin{aligned} T_{\text{roundT}} &= 2T_F + T_{\text{replyB}} \\ T_{\text{roundB}} &= 2T_F + T_{\text{replyT}}. \end{aligned} \tag{1}$$

Then, by multiplying T_{roundT} by T_{roundB} and rearranging, we can get

$$T_{\text{roundT}}T_{\text{roundB}} - T_{\text{replyT}}T_{\text{replyB}} = 2T_F\left(2T_F + T_{\text{replyT}} + T_{\text{replyB}}\right). \tag{2}$$

And by combining (1) with (2), the TOF can be obtained as follows:

$$T_F = \frac{T_{\text{roundT}} \times T_{\text{roundB}} - T_{\text{replyT}} \times T_{\text{replyB}}}{T_{\text{roundT}} + T_{\text{roundB}} + T_{\text{replyT}} + T_{\text{replyB}}}. \tag{3}$$

Then, multiplying the T_F by the speed of light c, the distance between tag and BS can be expressed as $d = cT_F$.

B. IMU Module

The inertial measurement unit (IMU) plays a crucial role in indoor positioning, providing essential data support for indoor navigation and attitude estimation. In our system, the IMU provides the direction and acceleration information. In order to appropriately describe the motion of pedestrian, it is important to project acceleration within the body coordinate system, as ascertained by the IMU, onto the local navigation frame. The local navigation coordinate is determined by one of anchor nodes directly. This decision is highly reasonable, as the BSs are in fixed positions.

Fig. 3 depicts the process of transforming coordinates from the body coordinate system ($Ox_1y_1z_1$) to the local navigation coordinate system ($Ox_3y_3z_3$). This conversion involves three essential rotations. Initially, there is a rotation by an angle r around the x_1-axis, resulting in the intermediate system denoted as $Ox_1y_2z_2$. Subsequently, a rotation by an angle p around the y_2-axis leads to the transformation to $Ox_2y_2z_3$. Finally, a rotation by an angle y around the z_3-axis brings the coordinates into the $Ox_3y_3z_3$. Each of these rotations is accompanied by a corresponding coordinate transformation matrix, specifically M_x, M_y, and M_z

$$\begin{aligned} \boldsymbol{M_x} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos r & \sin r \\ 0 & -\sin r & \cos r \end{bmatrix} \\ \boldsymbol{M_y} &= \begin{bmatrix} \cos p & 0 & -\sin p \\ 0 & 1 & 0 \\ \sin p & 0 & \cos p \end{bmatrix} \\ \boldsymbol{M_z} &= \begin{bmatrix} \cos y & \sin y & 0 \\ -\sin y & \cos y & 0 \\ 0 & 0 & 1 \end{bmatrix}. \end{aligned} \tag{4}$$

The rotation matrix from the coordinate system $Ox_1y_1z_1$ to $Ox_3y_3z_3$ can be obtained as follows:

$$\boldsymbol{C}_B^L = \boldsymbol{M}_z \cdot \boldsymbol{M}_y \cdot \boldsymbol{M}_x. \tag{5}$$

Consequently, it can derive the conversion equation that facilitates the projection of acceleration from the body coordinate system, denoted as $\boldsymbol{a}^B$, into the local navigation frame, represented as

$$\boldsymbol{a}^L = \boldsymbol{C}_B^L \cdot \boldsymbol{a}^B. \tag{6}$$

If we remove the gravitational acceleration g in the local navigation frame, the formula should be replaced by

$$\boldsymbol{a}^L = \boldsymbol{C}_B^L \boldsymbol{a}^B - \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix}. \tag{7}$$

III. Proposed Tracking Method

Particle filter (PF), also referred to as Monte Carlo filters, falls within the category of Bayesian filters utilized for estimating the current state of a dynamic system [29]. It represents a posterior distribution by maintaining a set of particles, with each corresponding to different state hypotheses. One compelling characteristic of sample-based filters is their ability

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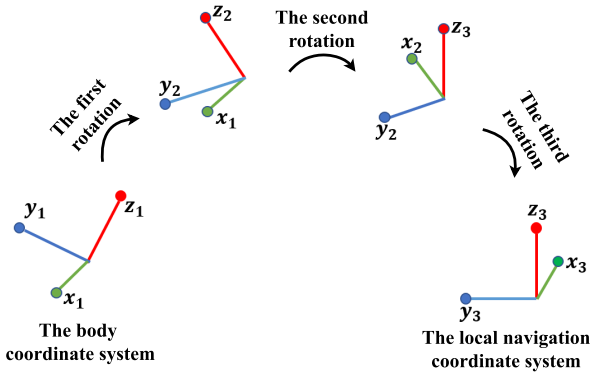


Fig. 3. Transformation of coordinate system.

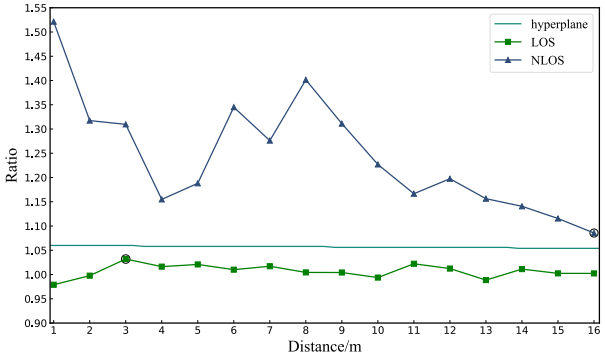


Fig. 4. Training results of the SVM model.

to approximate arbitrary distributions, a capability lacking in commonly used parametric models. However, traditional PF encounters challenges related to particle convergence and weak robustness. Therefore, in this article, we propose a DFRPF to fuse IMU and UWB for achieving high-accuracy positioning. The refined model accurately captures particles, while maintaining a reasonable variety of particles for a more precise approximation of the target distribution, resulting in reduced estimation error. It effectively addresses the challenges of particle convergence and weak robustness to NLOS. More details will be elaborated further below.

A. Motion Model

In a 2-D plane, we make the assumption that the tag moves with constant acceleration between two consecutive time stamps. The assumption is justified due to the typically short UWB sampling intervals. Therefore, the motion model for this case can be written as

$$\boldsymbol{X}_{k+1} = \boldsymbol{F}\boldsymbol{X}_k + \boldsymbol{B}\boldsymbol{U}_k + \boldsymbol{G}\boldsymbol{w}_k \tag{8}$$

where $\boldsymbol{X}_k$ represents the state vector at time k, which can be expressed as follows:

$$\boldsymbol{X}_k = \begin{bmatrix} x_x(k) & y_y(k) & v_x(k) & v_y(k) \end{bmatrix}^T \tag{9}$$

and

$$\boldsymbol{F} = \begin{bmatrix} 1 & 0 & t & 0 \\ 0 & 1 & 0 & t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix} \frac{t^2}{2} & 0 \\ 0 & \frac{t^2}{2} \\ t & 0 \\ 0 & t \end{bmatrix}$$

$$\boldsymbol{U}_k = \begin{bmatrix} a_x(k) \\ a_y(k) \end{bmatrix}, \quad \boldsymbol{G} = \begin{bmatrix} \frac{t^3}{6} & 0 \\ 0 & \frac{t^3}{6} \\ \frac{t^2}{2} & 0 \\ 0 & \frac{t^2}{2} \end{bmatrix}. \tag{10}$$

The sample time period is denoted as t. In this context, $\boldsymbol{F}$ represents the state transition matrix, $\boldsymbol{B}$ represents the control input matrix, and $\boldsymbol{U}_k$ is the control input vector. $\boldsymbol{G}$ is the noise driving matrix, $\boldsymbol{w}(k) = [\omega_x(k) \quad \omega_y(k)]^T$ represents the process noise vector with zero mean and a known covariance matrix $\boldsymbol{Q} = \text{diag}(\sigma_{a_x}^2, \sigma_{a_y}^2)$, which is essential in the observation model.

In our positioning system, the motion model is employed to generate the prior state $\boldsymbol{X}_{k|k-1}$ at time k based on the posterior state $\boldsymbol{X}_{k-1|k-1}$ at $k-1$. Subsequently, this prior state will be used in NLOS identification and particle sampling process.

B. SVM-Based NLOS Detection

The identification of NLOS has garnered significant attention in the field of UWB localization research. Many researchers have conducted extensive research on NLOS recognition, leveraging the potential of artificial intelligence theories [30], [31].

When in LOS condition, there exists an approximately linear relationship between the measured LOS ranging and the actual physical distance. However, the relationship becomes nonlinear due to various factors such as reflections and obstructions in NLOS condition. Based on this insight, we train a support vector machine (SVM) using ratio data extracted from ranging statistics. As for the training sample set, the position point is defined as

$$\left(\frac{d_{\text{nlos}}^i}{l_i}, +1\right), \quad i = 1, 2, \cdots, N \tag{11}$$

where d_{nlos}^i represents the ith NLOS measured value, N denotes the total number of measurements, and l_i denotes the actual distance. Additionally, we can calculate the negative point

$$\left(\frac{d_{\text{los}}^i}{l_i}, -1\right), \quad i = 1, 2, \cdots, N \tag{12}$$

by dividing the ith LOS measured value d_{los}^i by the actual distance l_i. This methodology forms the foundation of our training dataset, enabling further analysis and modeling. Utilizing the aforementioned dataset, the SVM model adeptly identifies a hyperplane tasked with effectively distinguishing between NLOS and LOS. The hyperplane is established through a linear system of equation denoted as

$$\boldsymbol{\omega}^T \boldsymbol{x} + \boldsymbol{b} = 0 \tag{13}$$

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where $\boldsymbol{b}$ represents the displacement term responsible for determining the spatial separation between the hyperplane and the origin, and $\boldsymbol{\omega}$ acts as the normal vector that denotes the orientation of the hyperplane. The training result is illustrated in Fig. 4. As shown, the SVM model exhibits remarkable capability in discerning NLOS instances from their LOS counterparts, thereby underscoring its capability in classification task.

Nevertheless, precise actual positions are often unattainable in practical scenarios. Therefore, we utilize the distance between the predicted state obtained from the motion model and the BS as the reference distance. Subsequently, this reference distance is combined with the UWB measurement distances to compute the input ratio.

C. DFRPF

1) Dynamic Feasible Region-Based Particle Sampling: During the prediction stage, the standard PF algorithm resamples particles after the state transition, potentially leading to particle convergence issues. To address the challenge, DFRPF incorporates the method of DFRPS for generating prior particles. The DFRPS establishes DFR based on the NLOS identification results and initializes the particle weight based on the angular difference at each prediction stage. It effectively addresses the problems of particle convergence and weak robustness to NLOS.

The establishment of DFR is related to the NLOS classification results. The identification of NLOS obtains N BSs in LOS condition and M BSs in NLOS condition. Given that NLOS distance measurements consistently exceed the actual distances, we apply a spatial constraint. Specifically, we restrict the potential target location to a circular region centered around the BS and with NLOS distance measurement as the radius. This constraint serves as the basis for defining the NLOS-based feasible area, denoted as

$$\boldsymbol{D}_1 = \begin{cases} \left\{\boldsymbol{x} \middle| \left\|\boldsymbol{x} - \boldsymbol{p}_{\text{nlos}}^i\right\|_2 \le r_k^i, i = 1, 2, \ldots, M\right\}, & M > 0 \\ \boldsymbol{R}^2, & M = 0 \end{cases} \tag{14}$$

where $\boldsymbol{x}$ represents the possible position of the target, and $\boldsymbol{p}_{\text{nlos}}^i$ denotes the position of the ith BS currently in NLOS condition. Meanwhile, r_k^i is the distance measured by the ith BS in NLOS condition. It is important to note that $\boldsymbol{R}^2$ encompasses the entirety of the 2-D space under consideration.

When BS in LOS state, it is observed that the ranging error adheres to a Gaussian distribution characterized by a mean (μ) and a standard deviation (σ) [32]. Leveraging the inherent properties of the Gaussian distribution, it can be deduced that the ranging error falls within the interval of $\mu - 2\sigma$ to $\mu + 2\sigma$ with a 95% confidence level. Consequently, we define the LOS-based feasible area as

$$\boldsymbol{D}_2 = \begin{cases} \left(\begin{cases} \boldsymbol{x} \middle| \hat{r}_k^j - \mu - 2\sigma \le \left\|\boldsymbol{x} - \boldsymbol{q}_{\text{los}}^j\right\|_2 \le \hat{r}_k^j \\ -\mu + 2\sigma, j = 1, 2, \ldots, N \end{cases}\right), & N > 0 \\ \boldsymbol{R}^2, & N = 0 \end{cases} \tag{15}$$

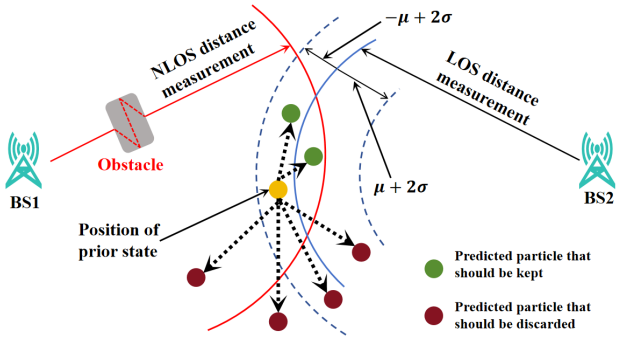


Fig. 5. Schematic of the particle generation method in a 2-D case.

where $\boldsymbol{x}$ represents the possible position of the target, and $\boldsymbol{q}_{\text{los}}^j$ and $\hat{r}_k^j$ denote the position and distance measurement of the jth BS under LOS condition, respectively. At last, by comprehensively considering both the LOS-based and NLOS-based feasible areas, we can obtain a DFR with enhanced robustness as $\boldsymbol{D} = \boldsymbol{D}_1 \cap \boldsymbol{D}_2$.

After determining the DFR, DFRPS starts to create a predicted particle by introducing Gaussian random process perturbations to the prior state $\boldsymbol{X}_{k|k-1}$. However, retaining the randomly generated particle only if its position falls within the $\boldsymbol{D}$ set. If the generated particle's position fails to satisfy this condition, it is promptly discarded, and repeating above process until a newly generated particle conforms to the DFR. This iterative approach forms a collection of particles $\boldsymbol{x}_k^{[1:n]} = \boldsymbol{x}_k^{[1]}, \ldots, \boldsymbol{x}_k^{[n]}$ at time k within the DFR, meeting the predefined particle number. For clarity, a schematic of the particle generation method in a 2-D case is illustrated in Fig. 5. As shown, the yellow circle represents the position of prior state, while the red circles represent the predicted particles that should be discarded. The green circles denote predicted particle that should been kept, since their position within the $\boldsymbol{D}$ set.

After obtaining the predicted particles, it becomes imperative to apply an evaluation function aimed at assessing the quality of each particle. In this context, we introduce an angle score denoted as

$$\text{ang}_i^j = \left|\arctan\left(\frac{y_k^{[i]} - y_b^j}{x_k^{[i]} - x_b^j}\right) - \arctan\left(\frac{y_{k|k-1} - y_b^j}{x_{k|k-1} - x_b^j}\right)\right| \tag{16}$$

where $(x_k^{[i]}, y_k^{[i]})$ stands for the coordinate of the ith particle $\boldsymbol{x}_k^{[i]}$ at time k, $(x_{k|k-1}, y_{k|k-1})$ is the coordinate of prior state at time k, and (x_b^j, y_b^j) represents the coordinate of the jth BS. This scoring mechanism is founded on the angular disparity observed between the angle of each particle in relation to the BSs and the angle of the prior state relative to the same BSs. As the angular disparity decreases, the corresponding angle score proportionally increases. However, it is worth noting that we have established a ceiling for the maximum attainable score, capping it at 500 to prevent unbounded values, as clearly

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defined below

$$\widehat{\omega}_k^{[i]} = \begin{cases} \dfrac{1}{\sum_{j=1}^{L} \text{ang}_i^j}, & \sum_{j=1}^{L} \text{ang}_i^j > 0.002 \\ 500, & \sum_{j=1}^{L} \text{ang}_i^j \leq 0.002 \end{cases} \tag{17}$$

where L is the total number of BS in the system. Then, after score normalization, the corresponding weight $\omega_k^{[i]}$ of the ith particle $\boldsymbol{x}_k^{[i]}$ can be initialized as

$$\omega_k^{[i]} = \frac{\widehat{\omega}_k^{[i]}}{\sum_{i=1}^{n} \widehat{\omega}_k^{[i]}}. \tag{18}$$

2) Importance Weighting: During the importance weighting stage, it is imperative to update the weight associated with each particle. In traditional particle filtering methods, the observation likelihood function remains static throughout the entire positioning process. This rigid condition will hinder the filter's ability to adapt to constantly changing environments. In our research, based on the fact that different UWB states have distinct ranging error models, we decompose the observation likelihood function into two distinct components to improve the adaptability and positioning accuracy of system across various situations.

As the ranging error model of UWB in LOS condition follows a Gaussian distribution [20], the importance weighting function can be defined as

$$p_{\text{los}}\left(z_k \middle| \boldsymbol{x}_k^{[i]}\right) = \begin{cases} \prod_{j=1}^{N} g_k\left(z_k^j \middle| \boldsymbol{x}_k^{[i]}\right), & N > 0 \\ 1, & N = 0 \end{cases} \tag{19}$$

where $g_k(z_k^j|\boldsymbol{x}_k^{[i]})$ represents the probability that the ith particle acquires the observed $\hat{d}_k^{ij}$ from the jth BS in LOS condition, which can be expressed as

$$g_k\left(z_k^j \middle| \boldsymbol{x}_k^{[i]}\right) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{\left(z_k^j - \hat{d}_k^{ij}\right)^2}{2\sigma^2}\right) \tag{20}$$

where $\hat{d}_k^{ij}$ is the distance between the jth BS and the particle $\boldsymbol{x}_k^{[i]}$ at step k. Additionally, z_k^j denotes the LOS measurement distance recorded by the jth BS. When there is no LOS measurements can be used, we set $p_{\text{los}}(z_k|\boldsymbol{x}_k^{[i]}) = 1$.

For NLOS measurement information, we employ a range error model following a Gamma distribution [15]. Therefore, the importance weighting function for NLOS condition can be defined as

$$p_{\text{nlos}}\left(z_k \middle| \boldsymbol{x}_k^{[i]}\right) = \begin{cases} \prod_{j=1}^{M} h_k\left(z_k^j \middle| \boldsymbol{x}_k^{[i]}\right), & M > 0 \\ 1, & M = 0 \end{cases} \tag{21}$$

where $h_k(z_k^j|\boldsymbol{x}_k^{[i]})$ represents the probability that the ith particle acquires the observed d_k^{ij} from the jth BS in NLOS condition, as defined below

$$h_k\left(z_k^j \middle| \boldsymbol{x}_k^{[i]}\right) = \alpha \cdot e^{-\alpha\left(z_k^j - d_k^{ij}\right)} \cdot \frac{\left(\alpha\left(z_k^j - d_k^{ij}\right)\right)^{\beta-1}}{\Gamma(\beta)} + c_0 \tag{22}$$

where α is the rate parameter and β represents the shape parameter characterizing the distribution. To deal with the issues of spurious measurements and ranging errors falling below zero, we thoughtfully introduce a constant term c_0. Additionally, d_k^{ij} is the distance between the jth BS and the particle $\boldsymbol{x}_k^{[i]}$ at step k, and z_k^j denotes the NLOS measurement distance recorded by the jth BS. So, the observation likelihood function can be got by incorporating the two importance weighting function

$$p\left(z_k \middle| \boldsymbol{x}_k^{[i]}\right) = p_{\text{los}}\left(z_k \middle| \boldsymbol{x}_k^{[i]}\right) \cdot p_{\text{nlos}}\left(z_k \middle| \boldsymbol{x}_k^{[i]}\right). \tag{23}$$

Then, the corresponding weight of each particle can be updated by the robust observation likelihood function

$$\omega_k^{[i]} = \eta \omega_k^{[i]} p\left(z_k \middle| \boldsymbol{x}_k^{[i]}\right) \tag{24}$$

where η is a normalizer that ensures the weights sum up to 1.

After calculating the weight for each particle, they are ranked in descending order based on their weight. Following this ranking, the top 15% of the particles are chosen to constitute the optimal particle set, and their average is computed to derive the filtered output state. The complete procedure for the proposed DFRPF is outlined in Algorithm 1.

IV. Implementations and Evaluations

In this section, to validate the effectiveness of the proposed algorithm for indoor positioning, we conduct tests in two distinct experimental scenarios and compare our proposed algorithm with three different positioning methods: the standard PF, robust PF (RPF), and UWB-RF [33]. UWB-RF is an indoor localization system that integrates the random forest algorithm and UWB through a Kalman filter to mitigate the influence of NLOS on localization precision. The RPF is a derivative of DFRPF, but it does not incorporate the DFRPS algorithm.

A. Experimental Scenarios

We have conducted extensive experiments in two distinct experimental scenarios. The Scenario 1 is located in a laboratory on the 17th floor of an office building, while Scenario 2 is situated in the public area on the 11th floor of the same office building. As depicted in Fig. 6(a) and (b), the Scenario 1 represents a relatively simple and weak NLOS environment, with obstacles like desks installed. In contrast, Scenario 2 represents a more complex and dynamic indoor setting, featuring concrete walls, furniture and pedestrian traffic as shown in Fig. 6(c). Additionally, the hallway in Scenario 2 poses a particularly challenging environment due to significant signal propagation obstruction caused by concrete walls, resulting in a hard NLOS environment. These areas exhibit different floor layouts, diverse wireless environments, and distinct user behavior patterns. The four yellow squares BS_1–BS_4 represent the BSs in our experiment, positioned at a height of 2 m. The green and pink circles are the starting and ending points of all experimental routes, respectively.

Additionally, three representative experiments are conducted in these two scenarios, labeled as Experiment 1, Experiment 2 and Experiment 3, respectively. Specifically, Experiment 1

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Algorithm 1 DFRPF

Input: Previous time filtered output: $\boldsymbol{X}_{k-1|k-1}$, control: $\boldsymbol{u}_k$, observation: $\boldsymbol{z}_k$, covariance matrix: $\boldsymbol{V}_k$, predefined particle number: n

$\widehat{S}_k = \varnothing$, $Q_k = \varnothing$;
// Determined the prior state at time step k;
$\boldsymbol{X}_{k|k-1} \leftarrow$ according (8);
//Setting dynamic feasible region;
$\boldsymbol{D}_1 \leftarrow$ according (14);
$\boldsymbol{D}_2 \leftarrow$ according (15);
$\boldsymbol{D} = \boldsymbol{D}_1 \cap \boldsymbol{D}_2$;
//Generate prior particles;
while $|\widehat{S}_k| < n$ **do**
  $\boldsymbol{x}_k^{[i]} \leftarrow \boldsymbol{X}_{k|k-1} + multivariate_normal(\boldsymbol{0}, \boldsymbol{V}_k)$;
  // $(x_k^{[i]}, y_k^{[i]})$ *is the coordinate of* $\boldsymbol{x}_k^{[i]}$;
  if $(x_k^{[i]}, y_k^{[i]}) \in \boldsymbol{D}$ **then**
    $\widehat{\omega}_k^{[i]} \leftarrow$ according (17);
    $\widehat{S}_k \cup \{< \boldsymbol{x}_k^{[i]}, \widehat{\omega}_k^{[i]} >\}$;
  else
    continue;
  end
end
foreach $\{< \boldsymbol{x}_k^{[i]}, \widehat{\omega}_k^{[i]} >\} \in \widehat{S}_k$ **do**
  $\omega_k^{[i]} = \frac{\widehat{\omega}_k^{[i]}}{\sum_{i=1}^{n} \widehat{\omega}_k^{[i]}}$;
end
//Observed likelihood function;
$p_{los}(\boldsymbol{z}_k|\boldsymbol{x}_k^{[i]}) \leftarrow$ according (19), (20);
$p_{nlos}(\boldsymbol{z}_k|\boldsymbol{x}_k^{[i]}) \leftarrow$ according (21), (22);
$p(\boldsymbol{z}_k|\boldsymbol{x}_k^{[i]}) = p_{los}(\boldsymbol{z}_k|\boldsymbol{x}_k^{[i]}) \cdot p_{nlos}(\boldsymbol{z}_k|\boldsymbol{x}_k^{[i]})$;
//Update weight for each particle;
forall $\{< \boldsymbol{x}_k^{[i]}, \omega_k^{[i]} >\} \in \widehat{S}_k$ **do**
  $\omega_k^{[i]} = \eta \omega_k^{[i]} p(\boldsymbol{z}_k|\boldsymbol{x}_k^{[i]})$;
end
$S_k \leftarrow sort(\widehat{S}_k)$ *//Sorted by the weight*;
//Save the particles of the first 15% *of the weight*;
for $i = 1$ **to** $n \cdot 15\%$ **do**
  $Q_k \cup \{\boldsymbol{x}_k^{[i]} | \boldsymbol{x}_k^{[i]} \in S_k\}$;
end
$\boldsymbol{X}_{k|k} \leftarrow mean(Q_k)$

and 2 are conducted in Scenario 1, while Experiment 3 takes place in Scenario 2, with each demonstrating distinct characteristics, as depicted in Fig. 6. In Experiment 1, volunteers walk along the preset trajectory at a constant speed, covering a total length of 20 m with only straight walking and right-angle turns. Experiment 2 features route with varying turning angles, including a 180° turn, covering a total distance of 32.82 m. The path of Experiment 3 needs to pass through a challenging NLOS area twice, covering a total distance of 48.95 m with multiple turns. During Experiment 2 and 3, the volunteers are required to randomly direct various behaviors, including normal walking, stepping on one place, and jogging at different speeds to make the experiment closer to the real indoor walking situation of pedestrian. Moreover, all the routes connect key points labeled on the floor as sampling points to evaluate the accuracy of positioning. When volunteers cross a key point, the timestamp is recorded, and then the position with timestamp closest to the key point is selected to quantify the position error. In our experiments, the paths in Experiment 1, 2, and 3 need to pass through 30, 50, and 80 key points, respectively.

B. Devices and Experimental Setup

In the experiments, four DWM1000 UWB nodes are configured as BSs, along with a DWM1000 UWB device designated as mobile target. Additionally, our experimental setup incorporates a six-axis IMU (MPU6050), which provides crucial acceleration and orientation data for our system. It is worth noting that, at the current stage of implementation, we choose an offline approach to process all relevant data from both the IMU and UWB after they have been recorded and saved.

The parameter configurations for the proposed DFRPF are crucial for the successful implementation of the algorithm. In this article, the Gaussian distribution is characterized by a mean value (μ) of 0.069 m and a standard deviation (σ) of 0.097 m. Meanwhile, the parameters α and β for the Gamma distribution are set to 3.5 and 2, respectively, in accordance with established practices in the field.

C. Experiments and Results

To quantitatively measure the accuracy of the positioning results, we employ the root-mean-square error (RMSE) as the primary evaluation metric. Furthermore, in addition to evaluating the accuracy of positioning, stability is also a crucial consideration for indoor positioning. Therefore, to assess the stability of the results, we examine the standard deviation associated with the localization data.

First, the tracking paths in Experiment 1 and 2, representing the estimated positions generated by these algorithms, are depicted in Fig. 7(a) and (b), respectively, with the black solid path indicating the reference trajectory. As shown, the UWB-RF exhibits the most pronounced performance oscillation. This shows that the impact of measurement noise from the UWB sensor nodes on indoor positioning accuracy. However, the sensor fusion-based methods effectively smooth and improve the trajectory results. And among the sensor fusion-based approaches, the trajectory produced by our proposed DFRPF algorithm is closer to the ground truth trajectory. Meanwhile, in order to quantitatively assess the localization performance of each algorithm, various perspectives, including average error, maximum value, and median error, are illustrated in Fig. 8 and Table I, with the error box plot depicted in Fig. 9(a) and (b). Within the box plot, the central line signifies the median positioning error, while the lower and upper edges of the box represent errors at the 25%–75%, respectively. From these figures, we can get the same conclusion that the absolute accuracy and stability of the system are all improved with the utilization of the sensor fusion-based approaches. Sensor fusion-based methods excel in suppressing errors by successfully integrating and mutually calibrating IMU and UWB, thereby reducing the positioning system's reliance on individual sensors. On the other hand, in contrast to

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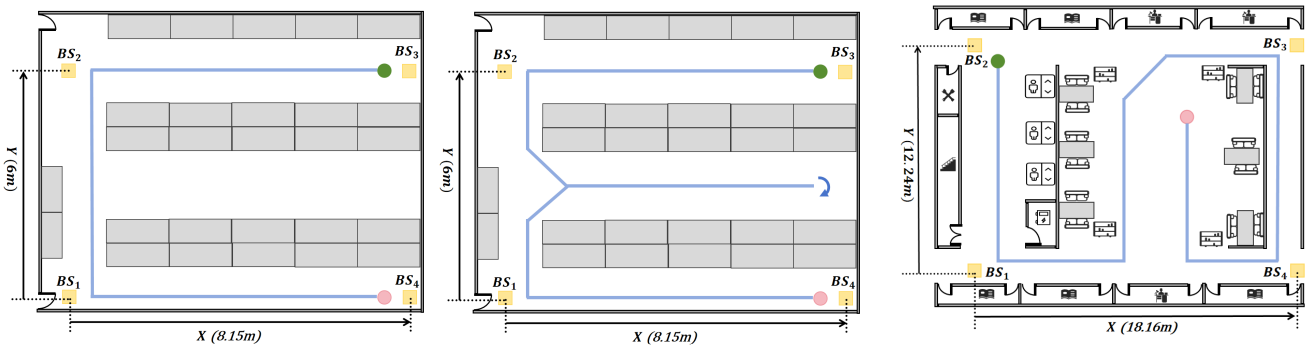


Fig. 6. Light blue lines are the reference trajectory. Yellow squares are BSs, while the green and pink circles mark the starting and ending points of all experimental paths, respectively. (a) Experiment 1. (b) Experiment 2. (c) Experiment 3.

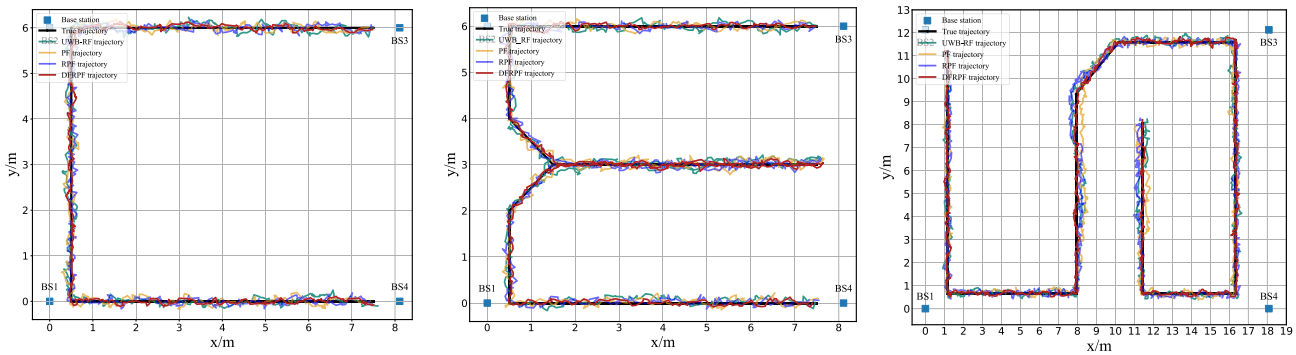


Fig. 7. Positioning trajectories generated by different algorithms. (a) Generated trajectories in Experiment 1. (b) Generated trajectories in Experiment 2. (c) Generated trajectories in Experiment 3.

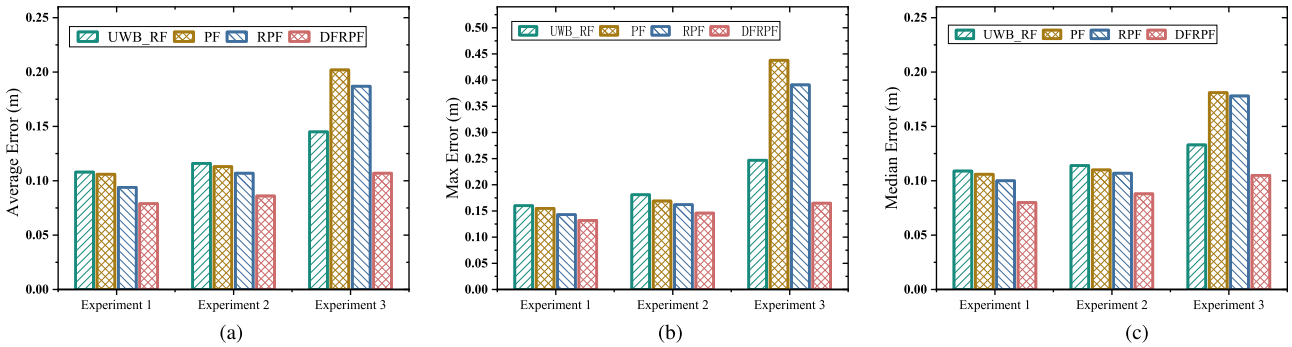


Fig. 8. Distinct perspectives of localization performance for these algorithms. (a) Average error in different experiments. (b) Maximum error in different experiments. (c) Median error in different experiments.

Experiment 1, Experiment 2 demands volunteers to randomly direct various behaviors, including normal walking, stepping on one place, and jogging at different speeds. This results in a certain degree of performance degradation across nearly all algorithms. Specifically, the average errors for UWB-RF, PF, and RPF have risen to 0.116, 0.113, and 0.107 m, respectively. Simultaneously, the maximum errors for UWB-RF, PF, and RPF have increased to 0.181, 0.169, and 0.162 m, respectively. However, the proposed DFRPF algorithm demonstrates resilience to these challenges and sustains a high level of performance, featuring a median error of 0.088 m, a maximum error of 0.146 m, and an average error of 0.086 m. This indicates that our algorithm has strong robustness and adaptability to diverse walking behaviors. Finally, when being focused on the stability of algorithm, the proposed DFRPF algorithm holds the best stability with the average standard deviation of 0.026 m in Experiment 1, outperforming RPF by 13.3%, PF by 18.8% and UWB-RF by 23.5%, respectively, and with the average standard deviation of 0.029 m in Experiment 2, outperforming RPF by 19.4%, PF by 25.6% and UWB-RF by 26.6%, respectively. The significant performance improvement observed with the DFRPF algorithm demonstrates that its stability across different situations.

In Experiment 3, the localization performance of these algorithms, similar to the tests above, has been illustrated through various perspectives in Figs. 8–9 and Table I. Certainly, there is a discernible decline in performance observed across most algorithms, with the RPF and PF algorithms notably affected by the challenging environment. The RPF demonstrates a maximum error and an average error of 0.391–0.187 m,

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TABLE I
COMPARISON OF LOCALIZATION PERFORMANCE UNDER DIFFERENT ALGORITHMS

Scenario number	Algorithm	Average error	Improved	Maximum error	Improved	Median error	Improved
Experiment 1	**DFRPF**	**0.079*m***	N/A	**0.132*m***	N/A	**0.080*m***	N/A
–	UWB-RF	$0.108m$	26.9%	$0.160m$	17.5%	$0.109m$	26.6%
–	PF	$0.106m$	25.5%	$0.155m$	14.8%	$0.106m$	24.5%
–	RPF	$0.094m$	16.0%	$0.143m$	7.69%	$0.100m$	20.0%
Experiment 2	**DFRPF**	**0.086*m***	N/A	**0.146*m***	N/A	**0.088*m***	N/A
–	UWB-RF	$0.116m$	25.9%	$0.181m$	19.3%	$0.114m$	22.8%
–	PF	$0.113m$	23.9%	$0.169m$	13.6%	$0.110m$	20.0%
–	RPF	$0.107m$	19.6%	$0.162m$	9.88%	$0.107m$	17.8%
Experiment 3	**DFRPF**	**0.107*m***	N/A	**0.165*m***	N/A	**0.105*m***	N/A
–	UWB-RF	$0.145m$	26.2%	$0.247m$	33.2%	$0.133m$	21.1%
–	PF	$0.202m$	47.0%	$0.438m$	62.3%	$0.181m$	42.0%
–	RPF	$0.187m$	42.8%	$0.391m$	57.8%	$0.178m$	41.0%

[1] The part in bold highlights the proposed algorithm and the corresponding good performance comparing with the others.
[2] The "Improved" column indicates the improvement achieved by the DFRPF comparing with the other filters.

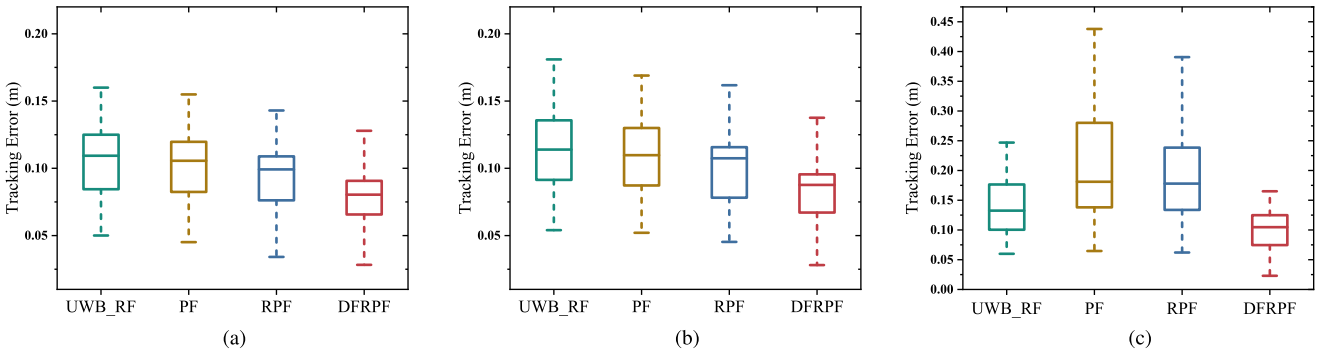


Fig. 9. Boxplots of error for different algorithms. (a) Boxplots of error in Experiment 1. (b) Boxplots of error in Experiment 2. (c) Boxplots of errors in Experiment 3.

respectively, while the PF exhibits slightly higher values with a maximum error of 0.438 m and an average error of 0.202 m, respectively. Notably, the trajectories generated by RPF and PF within the hall exhibit significant deviation from the reference path. This can be attributed to the heightened ranging error of UWB in challenging NLOS environments, posing difficulties in accurately correcting the cumulative error of IMU. As a result, sensor fusion failure occurs, thereby impacting the system's accuracy and reliability. This suggests that both the RPF and PF algorithms exhibit a strong reliance on the environment and demonstrate limited adaptability. Furthermore, UWB-RF outperforms RPF and PF by utilizing random forest methods to identify NLOS situations and dynamically adjusting the measurement distance. This capability effectively mitigates the unlimited degradation of positioning errors caused by NLOS condition. When focusing on the proposed DFRPF algorithm, it also exhibits high performance, with a median error of 0.105 m, and an average error of 0.107 m. The average positioning accuracy of the DFRPF algorithm is 42.8% higher than that of the RPF, 47.0% higher than the PF algorithm and 26.2% higher than the UWB-RF algorithm. Furthermore, as shown in Fig. 10(c), the proposed DFRPF algorithm holds the best stability with the average standard deviation of 0.035 m, outperforming RPF by 43.2%, PF by 49.7% and UWB-RF by 25.1%. Indeed, the DFRPF eliminates outliers and optimizes particle distribution with the DFR. This feature facilitates effective fusion of IMU and UWB in dynamic and complex environments, thereby increasing the proposed system's robustness and applicability.

When comparing RPF with PF in all tracking tests, the approach, dynamically adjusting the observation likelihood function based on the received UWB signal power quality, receives effective evaluation. The experimental results clearly indicate that RPF outperforms PF across all metrics, which demonstrates the approach effectively enhances the system's stability and positioning accuracy. Moreover, a thorough comparison between DFRPF and RPF algorithms in all tracking experiments provides a robust assessment of the DFR-based particle sampling (DFRPS) method. The average positioning accuracy, performance oscillation, median error and maximum error of the DFRPF outperform the RPF in all tracking tests. Especially in Experiment 3, the trajectory

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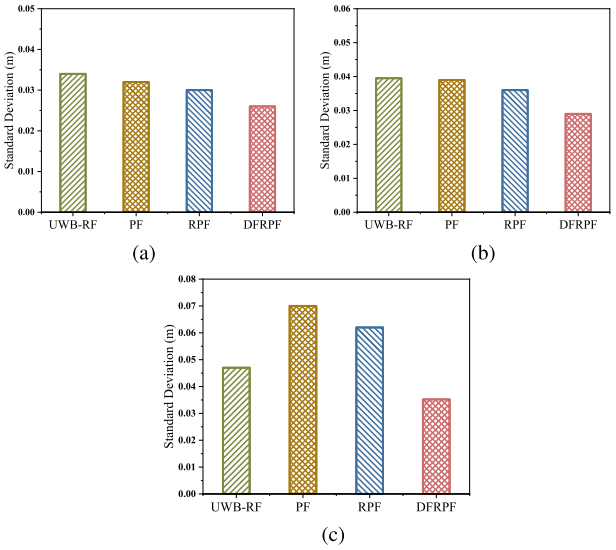


Fig. 10. Standard deviation of error under different algorithms. (a) Standard deviation of error in Experiment 1. (b) Standard deviation of error in Experiment 2. (c) Standard deviation of error in Experiment 3.

generated by RPF in the lobby exhibits significant deviation from the reference path, while the trajectory generated by DFRPF effectively follows the reference path. These experimental results indicate that DFRPS effectively reduces the system's dependence on the environment, making it more adaptable and stable across dynamic and complex environments. When comparing UWB-RF with DFRPF approach, our algorithm also outperforms UWB-RF in all experiments, highlighting its superior performance in managing NLOS situations.

Combining the results from all the tests allows us to draw the following conclusion that our proposed algorithm consistently exhibits significantly improved performance, maintaining high levels of accuracy and stability across different environments. Moreover, it effectively facilitates IMU and UWB fusion in dynamic and complex environments, reducing the impact of NLOS on system stability and accuracy, and enhancing the robustness and applicability of the system.

V. Conclusion

In this article, we propose a DFRPF, which incorporates a DFRPS methodology. Instead of conducting particle sampling following the state transition, the DFRPS adaptively defines a DFR based on the NLOS identification results for particle generation. In addition, to improve the adaptability of system across various situations, DFRPF dynamically adjusts the likelihood function according to different UWB states. In comparison to UWB-RF, PF, and RPF across all tracking scenarios, the proposed DFRPF algorithm exhibits the best performance, with an average error around 0.090 m, a maximum error around 0.148 m, a median error around 0.091 m, and an average standard deviation of localization error around 0.030 m. The experimental results demonstrate that the proposed DFRPF algorithm has the ability to handle the dynamic and complex indoor environments to realize a long-term reliability of positioning.

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